1. Load a Convolutional Neural Network

```
from torchvision.models import resnet50
import torch.nn as nn
import torch

# New weights with accuracy 80.858%

model = resnet50()
model.fc = nn.Linear(2048, len(classes))
model.load_state_dict(torch.load('./model.pt'))
```

```
<All keys matched successfully>
```

2. Test the network with an image

```
import torchvision.transforms.v2 as transforms
import torch

transform = transforms.Compose([
    transforms.Resize((256, 256)),
    transforms.ToImage(),  # Convert to image format
    transforms.ConvertImageDtype(torch.float32),  # Convert image dtype
    transforms.Normalize(mean=(0.5, 0.5, 0.5), std=(0.5, 0.5, 0.5))])
```

```
from PIL import Image



img_path = 'FruitClassification/test/kiwi/img_141.jpeg'
image = Image.open(img_path).convert('RGB')
image = transform(image)

# Check if GPU (cuda) is available
device = torch.device('cuda:0' if torch.cuda.is_available() else 'cpu')

# Move our model to device (GPU or CPU)
model.to(device)
model.eval()

image = image.to(device)
image = image.unsqueeze(0)

outputs = model(image)
outputs = outputs.squeeze(0)

score = torch.softmax(outputs, 0)
_, predicted = torch.max(score, 0)

print(f'Predicted: {classes[predicted]} {score[predicted] * 100:.02f}%')
```

```
Predicted: kiwi 99.89%
```